EXPERIMENT 37

Authentication Password-Failure Analysis

AIM

To identify repeated password failures against one account and failures distributed across multiple accounts from the same source.

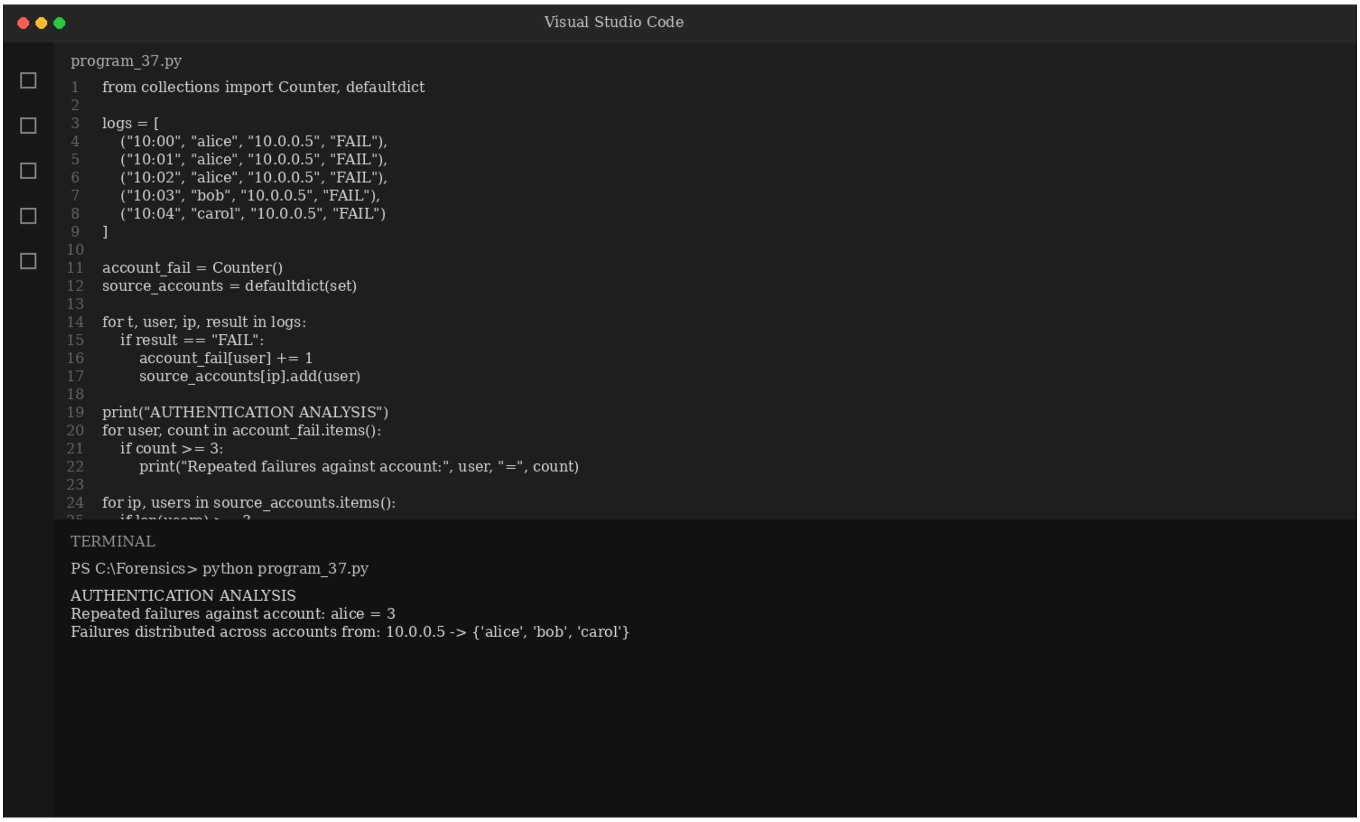
ALGORITHM

1. Read authentication records.
2. Count failures by username and source IP.
3. Detect repeated failures against one account.
4. Detect one source attacking multiple accounts.
5. Display separate findings.

CODE

```
from collections import Counter, defaultdict

logs = [
    ("10:00", "alice", "10.0.0.5", "FAIL"),
    ("10:01", "alice", "10.0.0.5", "FAIL"),
    ("10:02", "alice", "10.0.0.5", "FAIL"),
    ("10:03", "bob", "10.0.0.5", "FAIL"),
    ("10:04", "carol", "10.0.0.5", "FAIL")
]

account_fail = Counter()
source_accounts = defaultdict(set)

for t, user, ip, result in logs:
    if result == "FAIL":
        account_fail[user] += 1
        source_accounts[ip].add(user)

print("AUTHENTICATION ANALYSIS")
for user, count in account_fail.items():
    if count >= 3:
        print("Repeated failures against account:", user, "=", count)

for ip, users in source_accounts.items():
    if len(users) >= 3:
        print("Failures distributed across accounts from:", ip, "->", users)
```

OUTPUT



RESULT

The program identifies repeated authentication failures and failures distributed across multiple accounts.